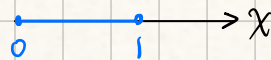
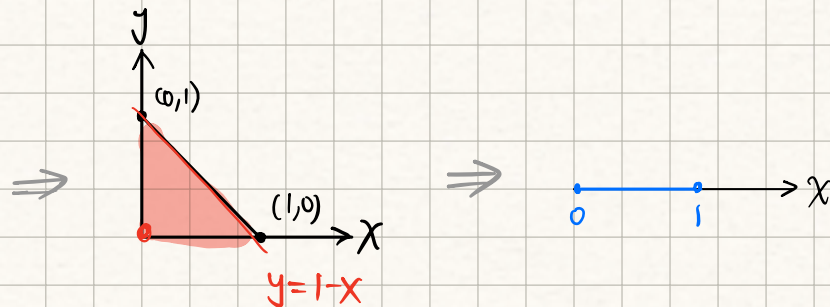
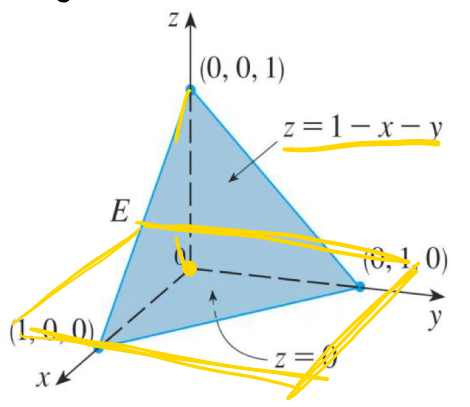


1.



$$\int_0^1 \int_0^{1-x} \int_0^{1-x-y} z \, dz \, dy \, dx = \int_0^1 \int_0^{1-x} \frac{1}{2}(1-x-y)^2 \, dy \, dx$$

$$= \frac{1}{2} \int_0^1 \frac{1}{3}(1-x)^3 \, dx$$

$$= \frac{1}{6} \int_0^1 v^3 \, dv$$

$$= \frac{1}{6} \left[\frac{1}{4} v^4 \right]_0^1$$

$$= \frac{1}{24} \quad \times$$

$$\int_0^{1-x-y} z \, dz = \left[\frac{1}{2} z^2 \right]_0^{1-x-y} = \frac{1}{2}(1-x-y)^2$$

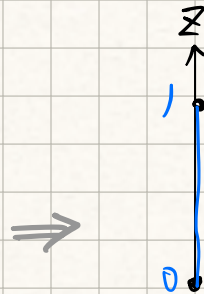
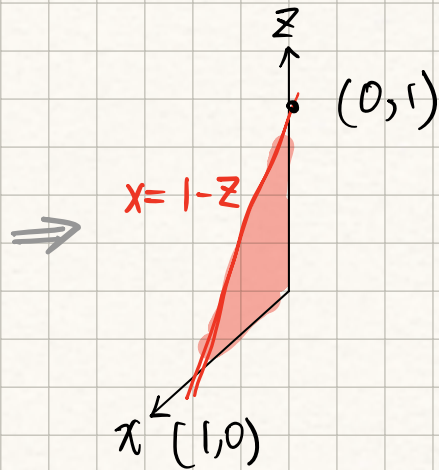
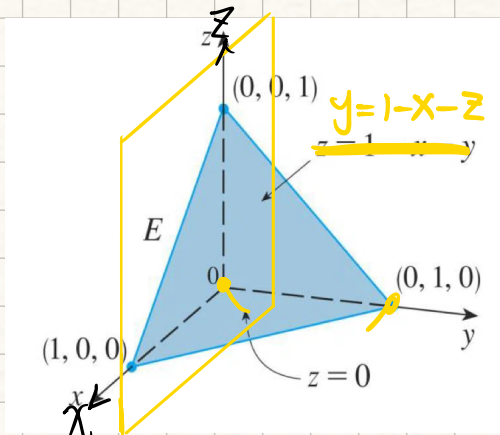
$$\int_0^a (a-y)^2 \, dy = -\int_0^a u^2 \, du$$

$$u = a - y \\ du = -dy$$

$$= -\frac{1}{3} a^3$$

$$= -\frac{1}{3} (1-x)^3$$

2.



$$\int_0^1 \int_0^{1-z} \int_0^{1-x-z} z \, dy \, dx \, dz = \int_0^1 \int_0^{1-z} z(1-x-z) \, dx \, dz$$

$$= \int_0^1 \frac{1}{2} z(1-z)^2 \, dz$$

$$= \frac{1}{2} \int_0^1 z - 2z^2 + z^3 \, dz$$

$$= \frac{1}{2} \cdot \left[\frac{1}{2} z^2 - \frac{2}{3} z^3 + \frac{1}{4} z^4 \right]_0^1$$

$$= \frac{1}{2} \cdot \left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4} \right) = \frac{1}{2} \cdot \frac{1}{12} = \frac{1}{24}$$

$$\alpha: \int_0^{1-x-z} z \, dy = \left[zy \right]_0^{1-x-z} = z(1-x-z)$$

$$\beta: \int_0^{1-z} z(1-x-z) \, dx = \left[z(1-z)x - \frac{1}{2} z x^2 \right]_0^{1-z}$$

$$= \left(z - \frac{1}{2} z \right) \cdot (1-z)^2$$

$$= \frac{1}{2} z(1-z)^2$$

$$z - zx - z^2$$

$$\Rightarrow zx - \frac{1}{2} zx^2 - z^2 x$$

$$= z(1-z)x - \frac{1}{2} zx^2$$